Project 5

Deploying and Managing Microservices in a Cloud-Native Environment

Project Title: Deploying and Managing Microservices in a Cloud-Native Environment

Objective: Containerize and deploy microservices using Kubernetes best practices, including deployments, services, autoscaling, and persistent storage.

1. Introduction

This project demonstrates how to deploy and manage a microservices-based application using Docker and Kubernetes. The system includes multiple services such as Product Service and Order Service, containerized using Docker and deployed to a Kubernetes cluster using YAML manifests.

2. Tools and Technologies Used

- Docker
- Kubernetes
- Minikube
- kubectl
- Docker Hub
- YAML
- Microservices Architecture
- Horizontal Pod Autoscaler (HPA)
- Persistent Volumes (PV) and Persistent Volume Claims (PVC)

3. Kubernetes Cluster Setup

Step 1: Install Docker Desktop.
Step 2: Install Minikube.
Step 3: Install kubectl.
Step 4: Start Minikube using command:
minikube start

Step 5: Verify cluster:
kubectl get nodes

Step 6: Verify pods:
kubectl get pods -A

4. Containerization using Docker

Created separate Dockerfiles for Product Service and Order Service.

Example Docker build command:
docker build -t username/product-service:v1 .

Push image to Docker Hub:
docker push username/product-service:v1

5. Kubernetes Deployments

Created deployment YAML files for each microservice.

Deployment includes:
- Container image
- Replicas
- Port mapping
- Environment variables

Deploy command:
kubectl apply -f deployment.yaml

6. Kubernetes Services

Created Kubernetes Service resources for communication.

- ClusterIP for internal communication
- NodePort for external access

Service deployment command:
kubectl apply -f service.yaml

Verify services:
kubectl get svc

7. Horizontal Pod Autoscaling (HPA)

Configured HPA for Product Service.

Command used:
kubectl autoscale deployment product-deployment --cpu-percent=50 --min=1 --max=5

Check autoscaler:
kubectl get hpa

Generated load using curl loop or hey tool to observe scaling.

8. Persistent Volume and Persistent Volume Claim

Created Persistent Volume (PV) and Persistent Volume Claim (PVC).

Order Service was modified to store and retrieve data using persistent storage.

Commands:
kubectl apply -f pv.yaml
kubectl apply -f pvc.yaml

Verified persistence after restarting pods.

9. Testing and Verification

Verified:
- Pod status
- Service communication
- External access
- Autoscaling
- Persistent data retention

10. Challenges Faced and Solutions

- Image pull errors solved by checking Docker Hub repository and tags.
- Pod CrashLoopBackOff fixed by checking container logs.
- HPA not scaling initially because metrics-server was missing.
- PVC binding issues resolved by matching storage class and access modes.

11. Conclusion

This project successfully demonstrated deployment and management of microservices in a cloud-native Kubernetes environment. Kubernetes features such as scaling, service discovery, and persistent storage were implemented successfully.

12. Sample Kubernetes YAML Files

Deployment YAML

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: product-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: product-service
  template:
    metadata:
      labels:
        app: product-service
    spec:
```

```yaml
      containers:
      - name: product-service
        image: username/product-service:v1
        ports:
        - containerPort: 8080
```

Service YAML

```yaml
apiVersion: v1
kind: Service
metadata:
  name: product-service
spec:
  type: NodePort
  selector:
    app: product-service
  ports:
  - port: 80
    targetPort: 8080
```

HPA YAML

```yaml
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: product-hpa
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: product-deployment
  minReplicas: 1
  maxReplicas: 5
  metrics:
  - type: Resource
    resource:
      name: cpu
      target:
        type: Utilization
        averageUtilization: 50
```

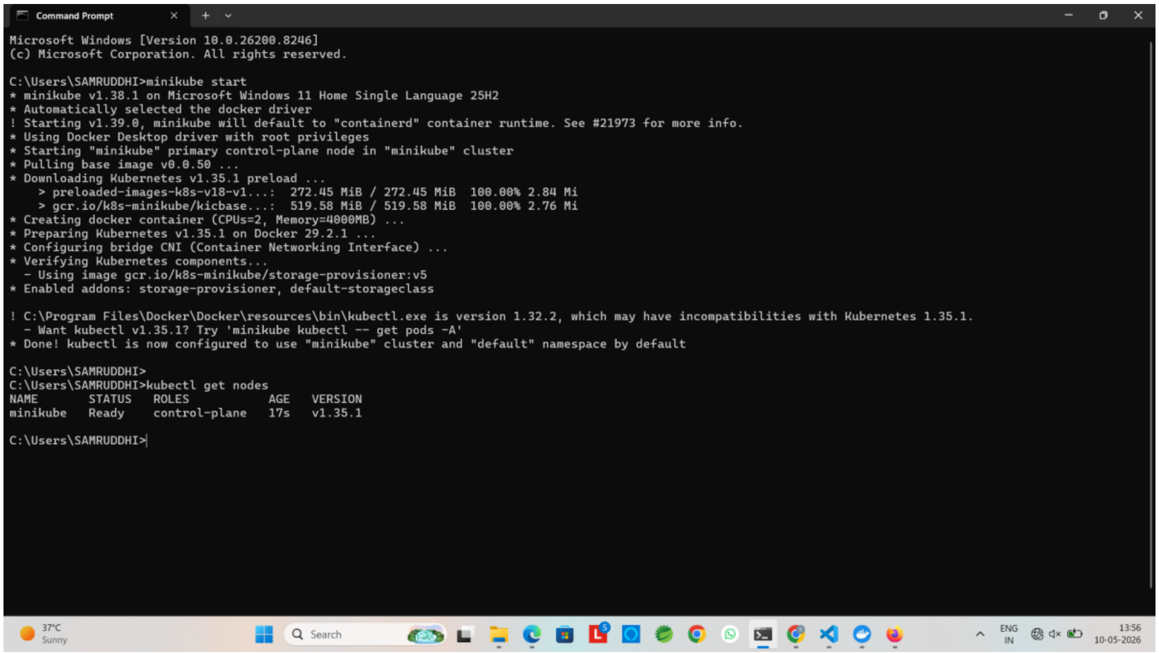
PV/PVC YAML

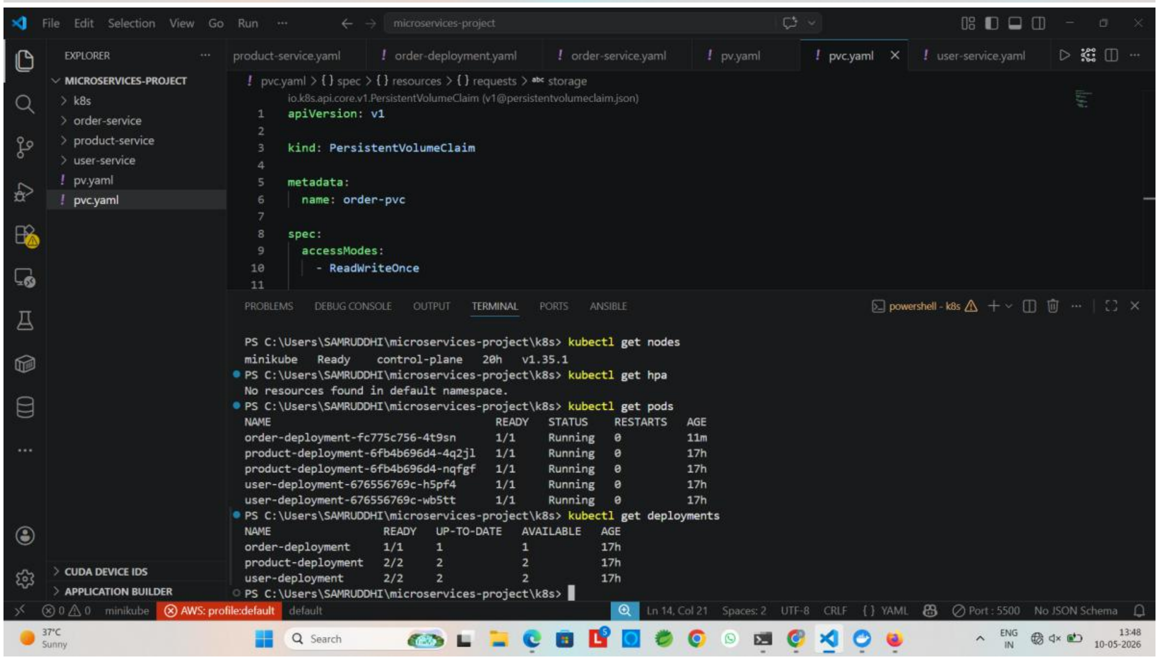
```yaml
apiVersion: v1
kind: PersistentVolume
metadata:
  name: app-pv
spec:
  capacity:
    storage: 1Gi
  accessModes:
    - ReadWriteOnce
  hostPath:
    path: "/data/app"

---
```

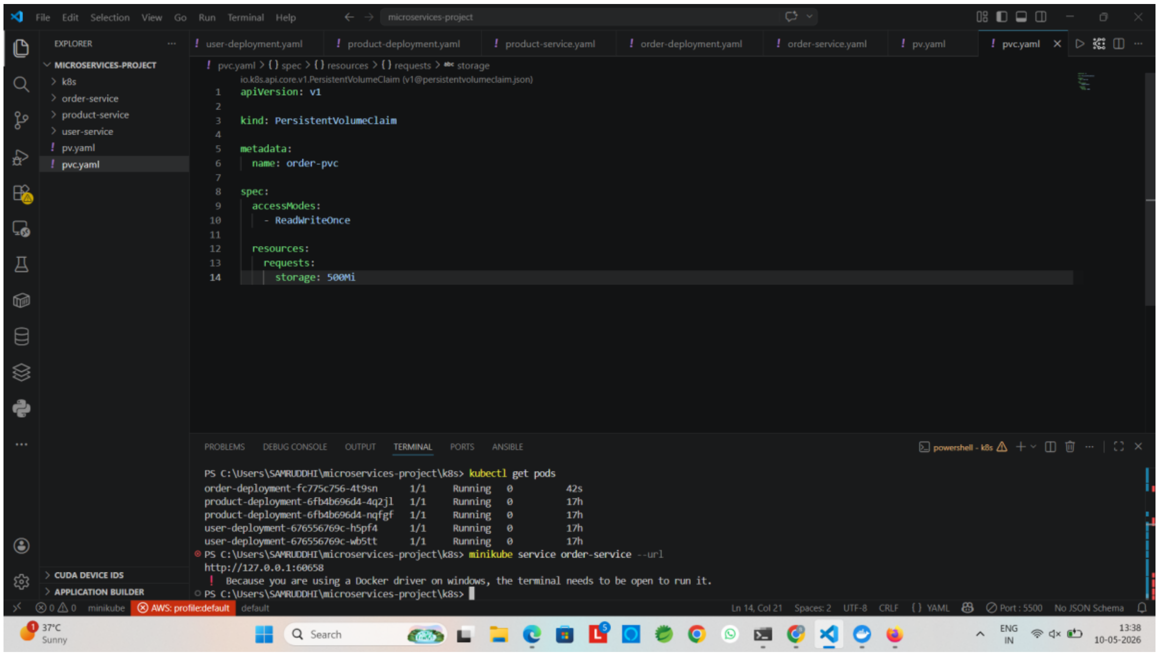
```yaml
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: app-pvc
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 1Gi
```

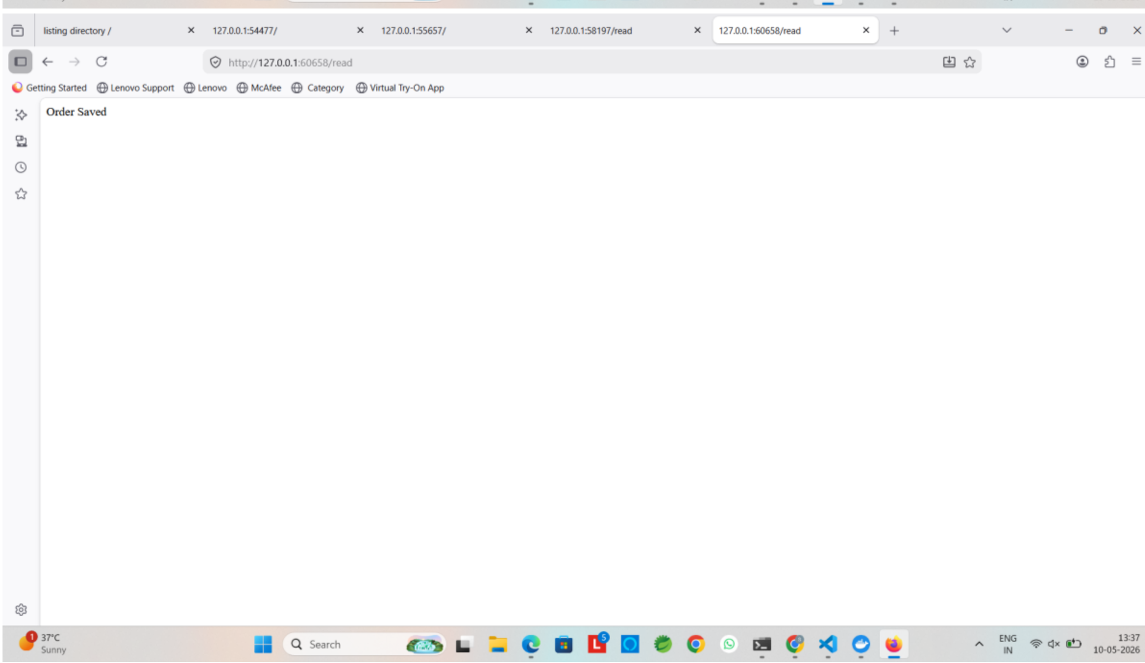
13. Screenshots to Add

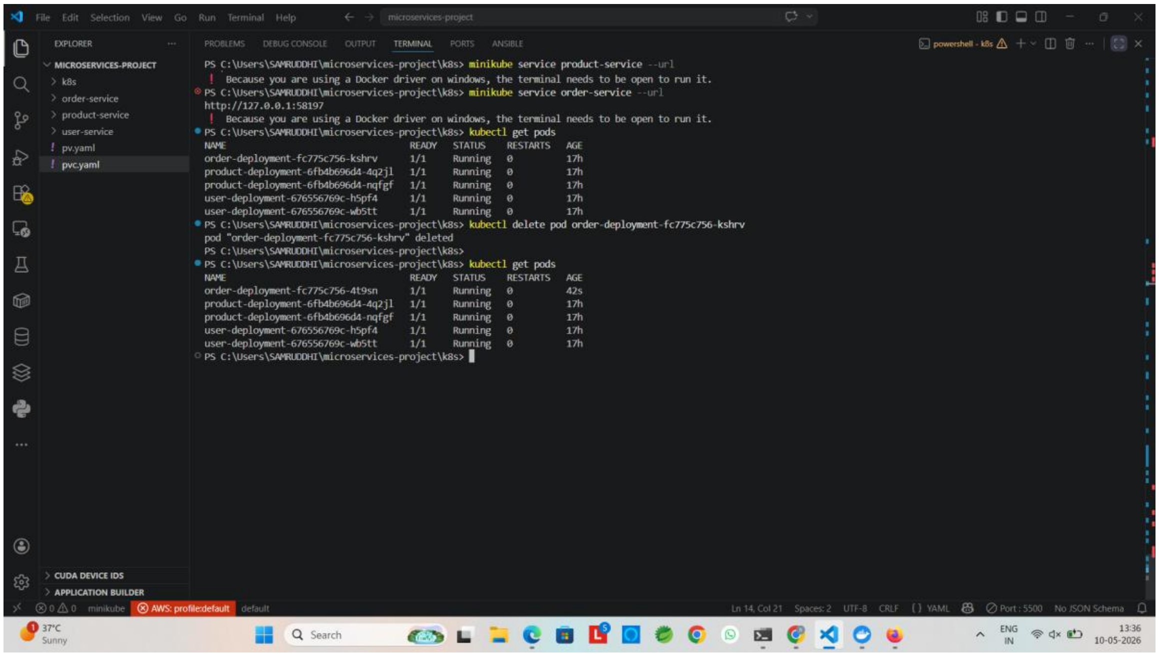
- Screenshot 1: Minikube cluster running
- Screenshot 2: kubectl get pods output
- Screenshot 3: Docker images built successfully
- Screenshot 4: Kubernetes deployments running
- Screenshot 5: Services exposed using NodePort
- Screenshot 6: HPA scaling output
- Screenshot 7: Persistent Volume and PVC status
- Screenshot 8: Data persistence after pod restart

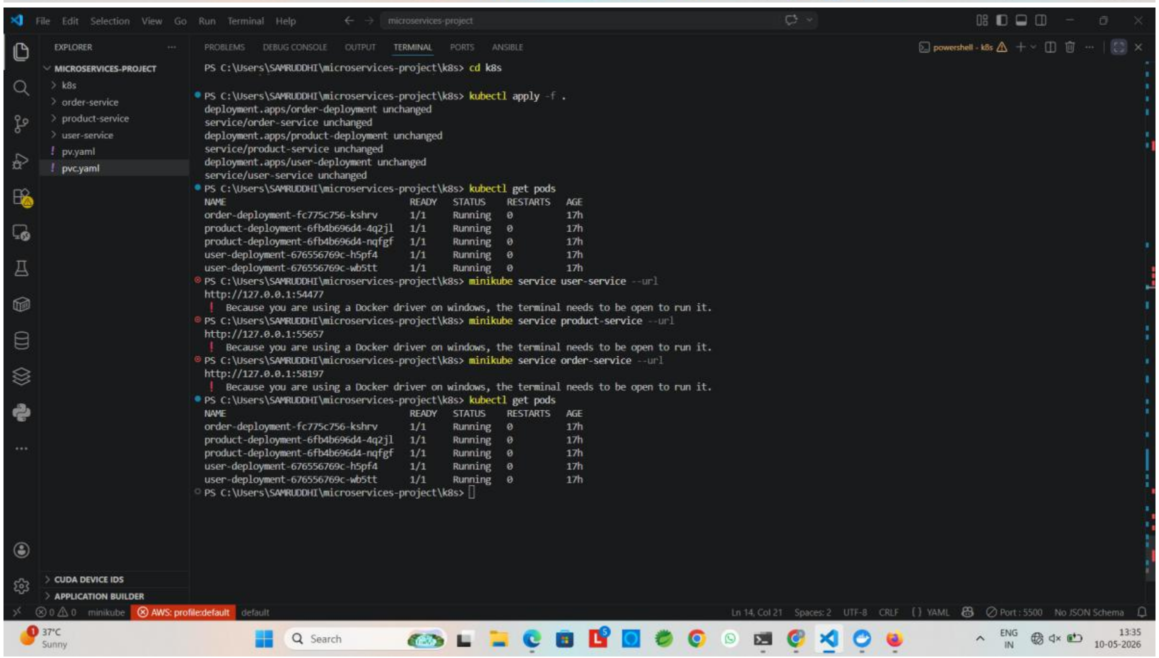

Command Prompt
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.
C:\Users\SAMRUDDHI>minikube start
* minikube v1.38.1 on Microsoft Windows 11 Home Single Language 25H2
* Automatically selected the docker driver
! Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
* Using Docker Desktop driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Downloading Kubernetes v1.35.1 preload ...
> preloaded-images-k8s-v18-v1...: 272.45 MiB / 272.45 MiB 100.00% 2.84 Mi
> gcr.io/k8s-minikube/kicbase...: 519.58 MiB / 519.58 MiB 100.00% 2.76 Mi
* Creating docker container (CPUs=2, Memory=4000MB) ...
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Configuring bridge CNI (Container Networking Interface) ...
* Verifying Kubernetes components...
- Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, default-storageclass
! C:\Program Files\Docker\Docker\resources\bin\kubectl.exe is version 1.32.2, which may have incompatibilities with Kubernetes 1.35.1.
- Want kubectl v1.35.1? Try 'minikube kubectl -- get pods -A'
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
C:\Users\SAMRUDDHI>
C:\Users\SAMRUDDHI>kubectl get nodes
NAME STATUS ROLES AGE VERSION
minikube Ready control-plane 17s v1.35.1
C:\Users\SAMRUDDHI>
37°C
Sunny
Search
ENG
IN
13:56
10-05-2026

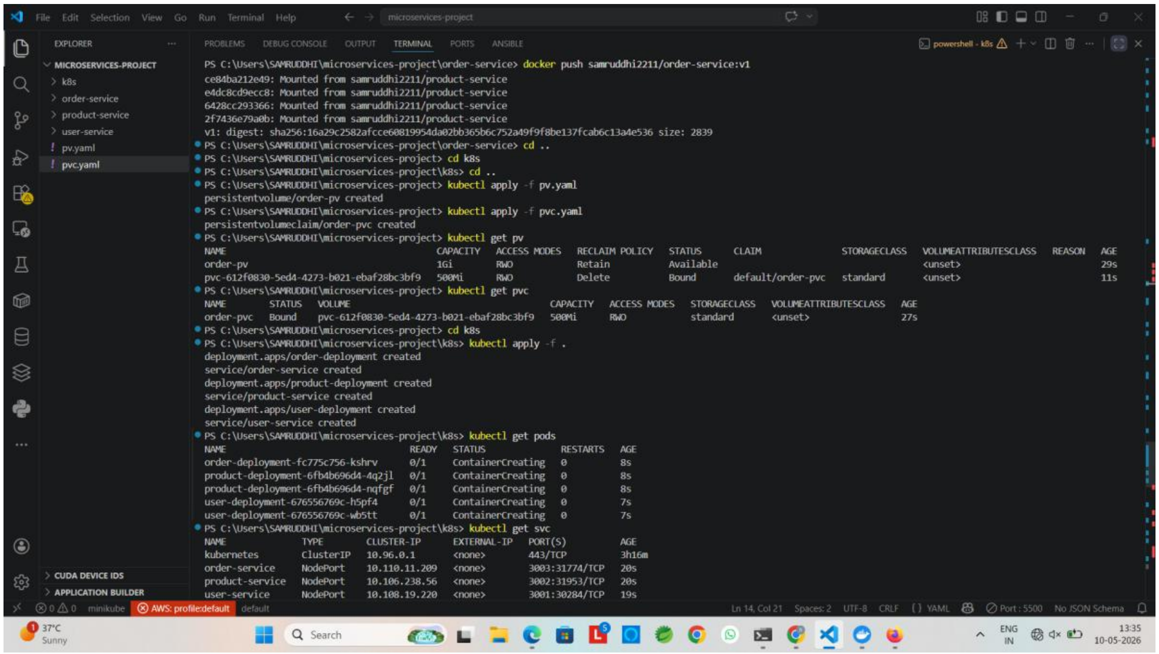

File Edit Selection View Go Run ...
microservices-project
EXPLORER
MICROSERVICES-PROJECT
k8s
order-service
product-service
user-service
pv.yaml
pvc.yaml
product-service.yaml
order-deployment.yaml
order-service.yaml
pv.yaml
pvc.yaml
user-service.yaml
pvc.yaml > {} spec > {} resources > {} requests > storage
io.k8s.api.core.v1.PersistentVolumeClaim (v1@persistentvolumeclaim.json)
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
name: order-pvc
spec:
accessModes:
- ReadWriteOnce
PROBLEMS DEBUG CONSOLE OUTPUT TERMINAL PORTS ANSIBLE
powershell - k8s
PS C:\Users\SAMRUDDHI\microservices-project\k8s> kubectl get nodes
minikube Ready control-plane 20h v1.35.1
PS C:\Users\SAMRUDDHI\microservices-project\k8s> kubectl get hpa
No resources found in default namespace.
PS C:\Users\SAMRUDDHI\microservices-project\k8s> kubectl get pods
NAME READY STATUS RESTARTS AGE
order-deployment-fc775c756-4t9sn 1/1 Running 0 11m
product-deployment-6fb4b696d4-4q2jl 1/1 Running 0 17h
product-deployment-6fb4b696d4-nqfgf 1/1 Running 0 17h
user-deployment-676556769c-h5pf4 1/1 Running 0 17h
user-deployment-676556769c-wb5tt 1/1 Running 0 17h
PS C:\Users\SAMRUDDHI\microservices-project\k8s> kubectl get deployments
NAME READY UP-TO-DATE AVAILABLE AGE
order-deployment 1/1 1 1 17h
product-deployment 2/2 2 2 17h
user-deployment 2/2 2 2 17h
PS C:\Users\SAMRUDDHI\microservices-project\k8s>
CUDA DEVICE IDS
APPLICATION BUILDER
minikube
AWS: profile:default
default
Ln 14, Col 21 Spaces: 2 UTF-8 CRLF {} YAML
Port : 5500 No JSON Schema
37°C
Sunny
Search
ENG
IN
13:48
10-05-2026


microservices-project
EXPLORER
MICROSERVICES-PROJECT
k8s
order-service
product-service
user-service
pv.yaml
pvc.yaml
user-deployment.yaml
product-deployment.yaml
product-service.yaml
order-deployment.yaml
order-service.yaml
pv.yaml
pvc.yaml
pvc.yaml > {} spec > {} resources > {} requests > storage
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
name: order-pvc
spec:
accessModes:
- ReadWriteOnce
resources:
requests:
storage: 500Mi
PROBLEMS
DEBUG CONSOLE
OUTPUT
TERMINAL
PORTS
ANSIBLE
PS C:\Users\SAMRUDDHI\microservices-project\k8s> kubectl get pods
order-deployment-fc775c756-4t9sn 1/1 Running 0 42s
product-deployment-6fb4b696d4-4q2jl 1/1 Running 0 17h
product-deployment-6fb4b696d4-nqfgf 1/1 Running 0 17h
user-deployment-676556769c-h5pf4 1/1 Running 0 17h
user-deployment-676556769c-wb5tt 1/1 Running 0 17h
PS C:\Users\SAMRUDDHI\microservices-project\k8s> minikube service order-service --url
http://127.0.0.1:60658
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
PS C:\Users\SAMRUDDHI\microservices-project\k8s>
CUDA DEVICE IDS
APPLICATION BUILDER
minikube
AWS: profile:default
Ln 14, Col 21 Spaces: 2 UTF-8 CRLF YAML
37°C
Sunny
Search
ENG
IN
13:38
10-05-2026

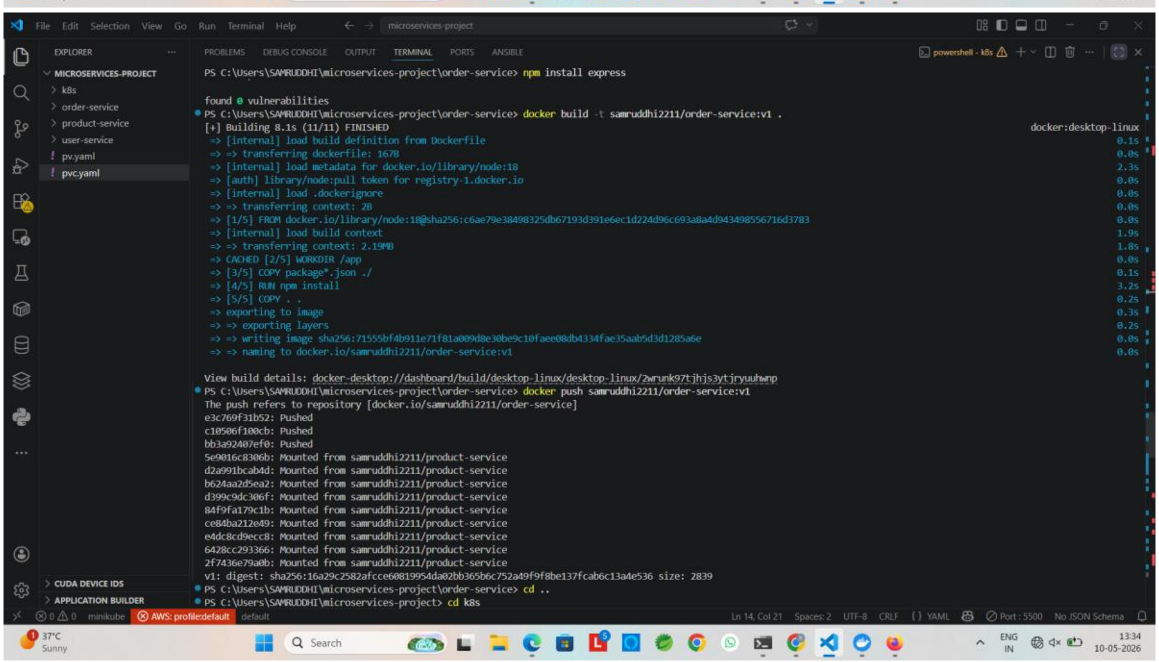

listing directory /
127.0.0.1:54477/
127.0.0.1:55657/
127.0.0.1:58197/read
127.0.0.1:60658/read
http://127.0.0.1:60658/read
Getting Started
Lenovo Support
Lenovo
McAfee
Category
Virtual Try-On App
Order Saved
37°C
Sunny
Search
ENG
IN
13:37
10-05-2026


PS C:\Users\SAMRUDDHI\microservices-project\k8s> minikube service product-service --url
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
PS C:\Users\SAMRUDDHI\microservices-project\k8s> minikube service order-service --url
http://127.0.0.1:58197
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
PS C:\Users\SAMRUDDHI\microservices-project\k8s> kubectl get pods
NAME READY STATUS RESTARTS AGE
order-deployment-fc775c756-kshrv 1/1 Running 0 17h
product-deployment-6fb4b696d4-4q2jl 1/1 Running 0 17h
product-deployment-6fb4b696d4-nqfgf 1/1 Running 0 17h
user-deployment-676556769c-h5pf4 1/1 Running 0 17h
user-deployment-676556769c-wb5tt 1/1 Running 0 17h
PS C:\Users\SAMRUDDHI\microservices-project\k8s> kubectl delete pod order-deployment-fc775c756-kshrv
pod "order-deployment-fc775c756-kshrv" deleted
PS C:\Users\SAMRUDDHI\microservices-project\k8s>
PS C:\Users\SAMRUDDHI\microservices-project\k8s> kubectl get pods
NAME READY STATUS RESTARTS AGE
order-deployment-fc775c756-4t9sn 1/1 Running 0 42s
product-deployment-6fb4b696d4-4q2jl 1/1 Running 0 17h
product-deployment-6fb4b696d4-nqfgf 1/1 Running 0 17h
user-deployment-676556769c-h5pf4 1/1 Running 0 17h
user-deployment-676556769c-wb5tt 1/1 Running 0 17h
PS C:\Users\SAMRUDDHI\microservices-project\k8s>

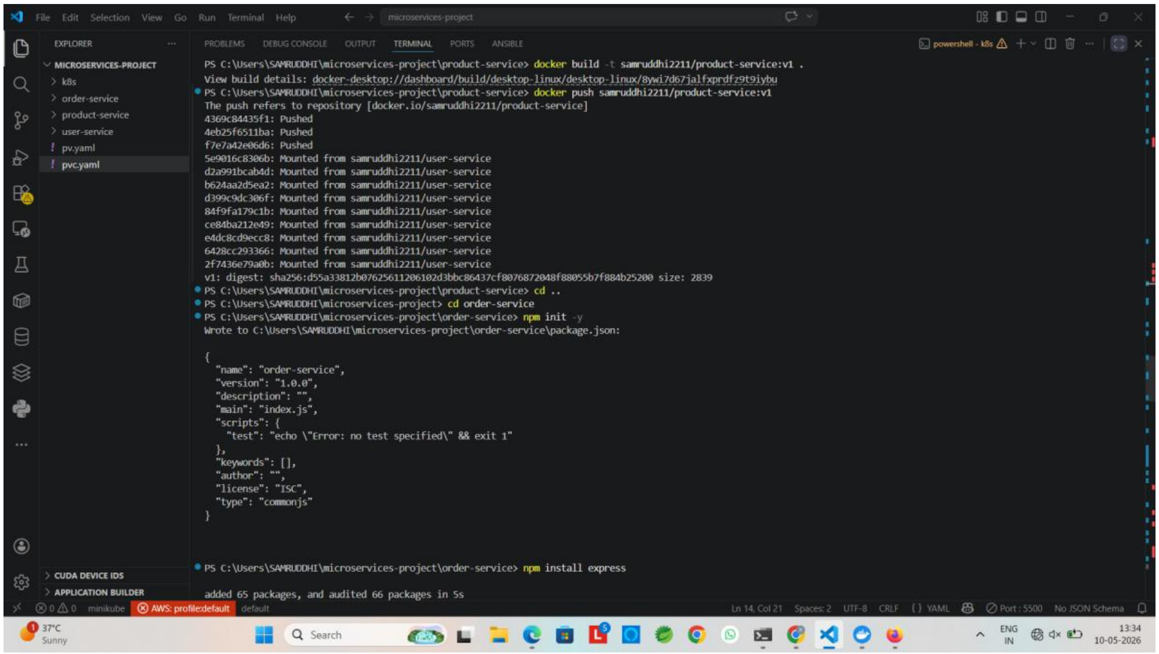

PS C:\Users\SAMRUDDHI\microservices-project\k8s> cd k8s
PS C:\Users\SAMRUDDHI\microservices-project\k8s> kubectl apply -f .
deployment.apps/order-deployment unchanged
service/order-service unchanged
deployment.apps/product-deployment unchanged
service/product-service unchanged
deployment.apps/user-deployment unchanged
service/user-service unchanged
PS C:\Users\SAMRUDDHI\microservices-project\k8s> kubectl get pods
NAME READY STATUS RESTARTS AGE
order-deployment-fc775c756-kshrv 1/1 Running 0 17h
product-deployment-6fb4b696d4-4q2jl 1/1 Running 0 17h
product-deployment-6fb4b696d4-nqfgf 1/1 Running 0 17h
user-deployment-676556769c-h5pf4 1/1 Running 0 17h
user-deployment-676556769c-wb5tt 1/1 Running 0 17h
PS C:\Users\SAMRUDDHI\microservices-project\k8s> minikube service user-service --url
http://127.0.0.1:54477
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
PS C:\Users\SAMRUDDHI\microservices-project\k8s> minikube service product-service --url
http://127.0.0.1:55657
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
PS C:\Users\SAMRUDDHI\microservices-project\k8s> minikube service order-service --url
http://127.0.0.1:58197
! Because you are using a Docker driver on windows, the terminal needs to be open to run it.
PS C:\Users\SAMRUDDHI\microservices-project\k8s> kubectl get pods
NAME READY STATUS RESTARTS AGE
order-deployment-fc775c756-kshrv 1/1 Running 0 17h
product-deployment-6fb4b696d4-4q2jl 1/1 Running 0 17h
product-deployment-6fb4b696d4-nqfgf 1/1 Running 0 17h
user-deployment-676556769c-h5pf4 1/1 Running 0 17h
user-deployment-676556769c-wb5tt 1/1 Running 0 17h
PS C:\Users\SAMRUDDHI\microservices-project\k8s>



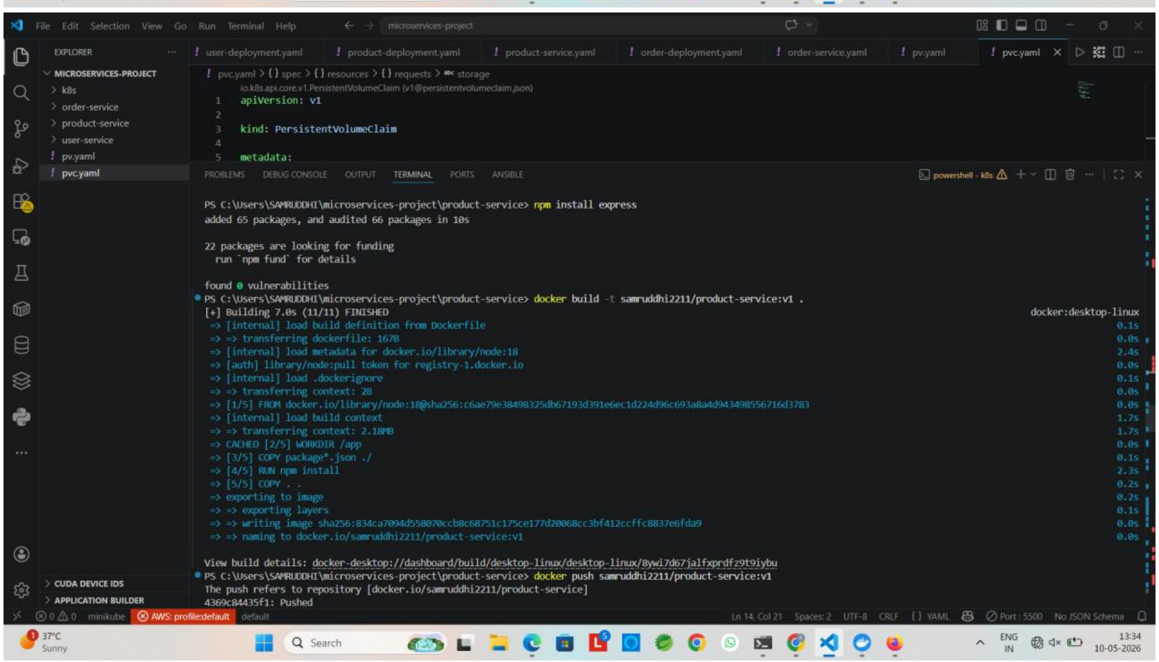
microservices-project
EXPLORER
MICROSERVICES-PROJECT
k8s
order-service
product-service
user-service
pv.yaml
pvc.yaml
PROBLEMS DEBUG CONSOLE OUTPUT TERMINAL PORTS ANSIBLE
powershell - k8s
PS C:\Users\SAMRUDDHI\microservices-project\order-service> docker push samruddhi2211/order-service:v1
ce84ba212e49: Mounted from samruddhi2211/product-service
e4dc8cd9ecc8: Mounted from samruddhi2211/product-service
6428cc293366: Mounted from samruddhi2211/product-service
2f7436e79a0b: Mounted from samruddhi2211/product-service
v1: digest: sha256:16a29c2582afcce60819954da02bb365b6c752a49f9f8be137fcab6c13a4e536 size: 2839
PS C:\Users\SAMRUDDHI\microservices-project\order-service> cd ..
PS C:\Users\SAMRUDDHI\microservices-project> cd k8s
PS C:\Users\SAMRUDDHI\microservices-project\k8s> cd ..
PS C:\Users\SAMRUDDHI\microservices-project> kubectl apply -f pv.yaml
persistentvolume/order-pv created
PS C:\Users\SAMRUDDHI\microservices-project> kubectl apply -f pvc.yaml
persistentvolumeclaim/order-pvc created
PS C:\Users\SAMRUDDHI\microservices-project> kubectl get pv
NAME CAPACITY ACCESS MODES RECLAIM POLICY STATUS CLAIM STORAGECLASS VOLUMEATTRIBUTESCLASS REASON AGE
order-pv 1Gi RWO Retain Available <unset> 29s
pvc-612f0830-5ed4-4273-b021-ebaf28bc3bf9 500Mi RWO Delete Bound default/order-pvc standard <unset> 11s
PS C:\Users\SAMRUDDHI\microservices-project> kubectl get pvc
NAME STATUS VOLUME CAPACITY ACCESS MODES STORAGECLASS VOLUMEATTRIBUTESCLASS AGE
order-pvc Bound pvc-612f0830-5ed4-4273-b021-ebaf28bc3bf9 500Mi RWO standard <unset> 27s
PS C:\Users\SAMRUDDHI\microservices-project> cd k8s
PS C:\Users\SAMRUDDHI\microservices-project\k8s> kubectl apply -f .
deployment.apps/order-deployment created
service/order-service created
deployment.apps/product-deployment created
service/product-service created
deployment.apps/user-deployment created
service/user-service created
PS C:\Users\SAMRUDDHI\microservices-project\k8s> kubectl get pods
NAME READY STATUS RESTARTS AGE
order-deployment-fc775c756-kshrv 0/1 ContainerCreating 0 8s
product-deployment-6fb4b696d4-4q2jl 0/1 ContainerCreating 0 8s
product-deployment-6fb4b696d4-nqfgf 0/1 ContainerCreating 0 8s
user-deployment-676556769c-h5pf4 0/1 ContainerCreating 0 7s
user-deployment-676556769c-wb5tt 0/1 ContainerCreating 0 7s
PS C:\Users\SAMRUDDHI\microservices-project\k8s> kubectl get svc
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
kubernetes ClusterIP 10.96.0.1 <none> 443/TCP 3h16m
order-service NodePort 10.110.11.209 <none> 3003:31774/TCP 20s
product-service NodePort 10.106.238.56 <none> 3002:31953/TCP 20s
user-service NodePort 10.108.19.220 <none> 3001:30284/TCP 19s
CUDA DEVICE IDS
APPLICATION BUILDER
minikube
AWS: profile:default
default
Ln 14, Col 21 Spaces: 2 UTF-8 CRLF { } YAML Port : 5500 No JSON Schema
37°C
Sunny
Search
ENG
IN
13:35
10-05-2026



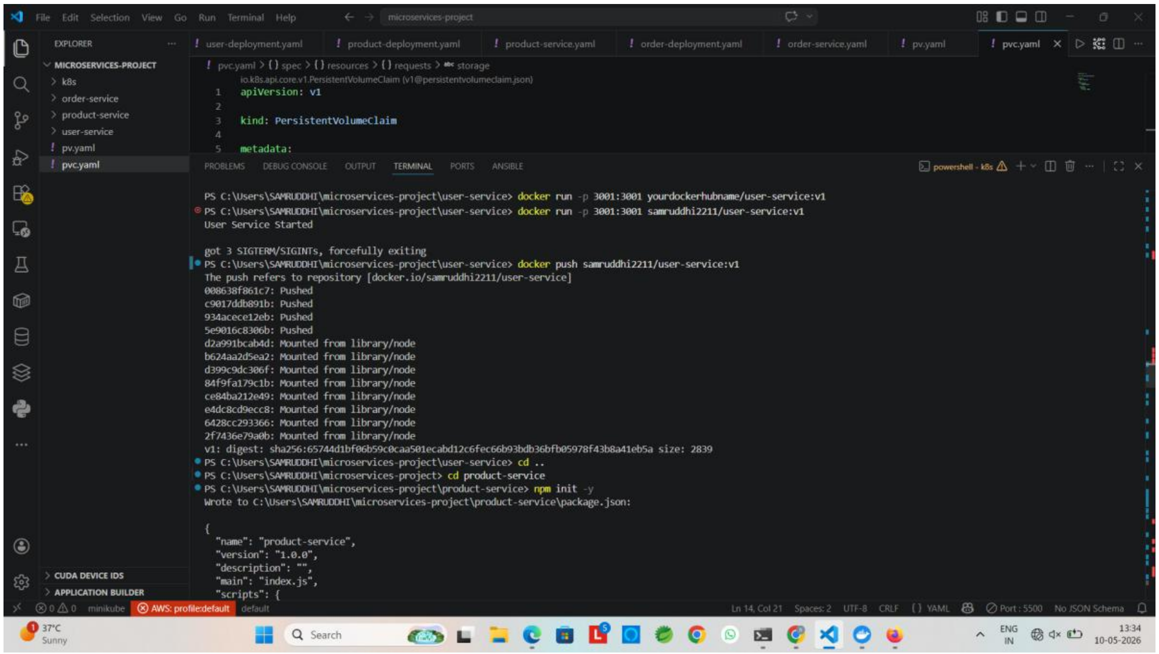
microservices-project
EXPLORER
MICROSERVICES-PROJECT
k8s
order-service
product-service
user-service
pv.yaml
pvc.yaml
PROBLEMS DEBUG CONSOLE OUTPUT TERMINAL PORTS ANSIBLE
powershell - k8s
PS C:\Users\SAMRUDDHI\microservices-project\order-service> npm install express
found 0 vulnerabilities
PS C:\Users\SAMRUDDHI\microservices-project\order-service> docker build -t samruddhi2211/order-service:v1 .
[+] Building 8.1s (11/11) FINISHED docker:desktop-linux
=> [internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 167B 0.0s
=> [internal] load metadata for docker.io/library/node:18 2.3s
=> [auth] library/node:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/5] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4d943498556716d3783 0.0s
=> [internal] load build context 1.9s
=> => transferring context: 2.19MB 1.8s
=> CACHED [2/5] WORKDIR /app 0.0s
=> [3/5] COPY package*.json ./ 0.1s
=> [4/5] RUN npm install 3.2s
=> [5/5] COPY . . 0.2s
=> exporting to image 0.3s
=> => exporting layers 0.2s
=> => writing image sha256:71555bf4b911e71f81a009d8e30be9c10faee08db4334fae35aab5d3d1285a6e 0.0s
=> => naming to docker.io/samruddhi2211/order-service:v1 0.0s
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/2wrunk97tjhjs3ytjryuuhwnp
PS C:\Users\SAMRUDDHI\microservices-project\order-service> docker push samruddhi2211/order-service:v1
The push refers to repository [docker.io/samruddhi2211/order-service]
e3c769f31b52: Pushed
c10506f100cb: Pushed
bb3a92407ef0: Pushed
5e9016c8306b: Mounted from samruddhi2211/product-service
d2a991bcab4d: Mounted from samruddhi2211/product-service
b624aa2d5ea2: Mounted from samruddhi2211/product-service
d399c9dc306f: Mounted from samruddhi2211/product-service
84f9fa179c1b: Mounted from samruddhi2211/product-service
ce84ba212e49: Mounted from samruddhi2211/product-service
e4dc8cd9ecc8: Mounted from samruddhi2211/product-service
6428cc293366: Mounted from samruddhi2211/product-service
2f7436e79a0b: Mounted from samruddhi2211/product-service
v1: digest: sha256:16a29c2582afcce60819954da02bb365b6c752a49f9f8be137fcab6c13a4e536 size: 2839
PS C:\Users\SAMRUDDHI\microservices-project\order-service> cd ..
PS C:\Users\SAMRUDDHI\microservices-project> cd k8s
CUDA DEVICE IDS
APPLICATION BUILDER
minikube
AWS: profile:default
default
Ln 14, Col 21 Spaces: 2 UTF-8 CRLF { } YAML Port : 5500 No JSON Schema
37°C
Sunny
Search
ENG
IN
13:34
10-05-2026



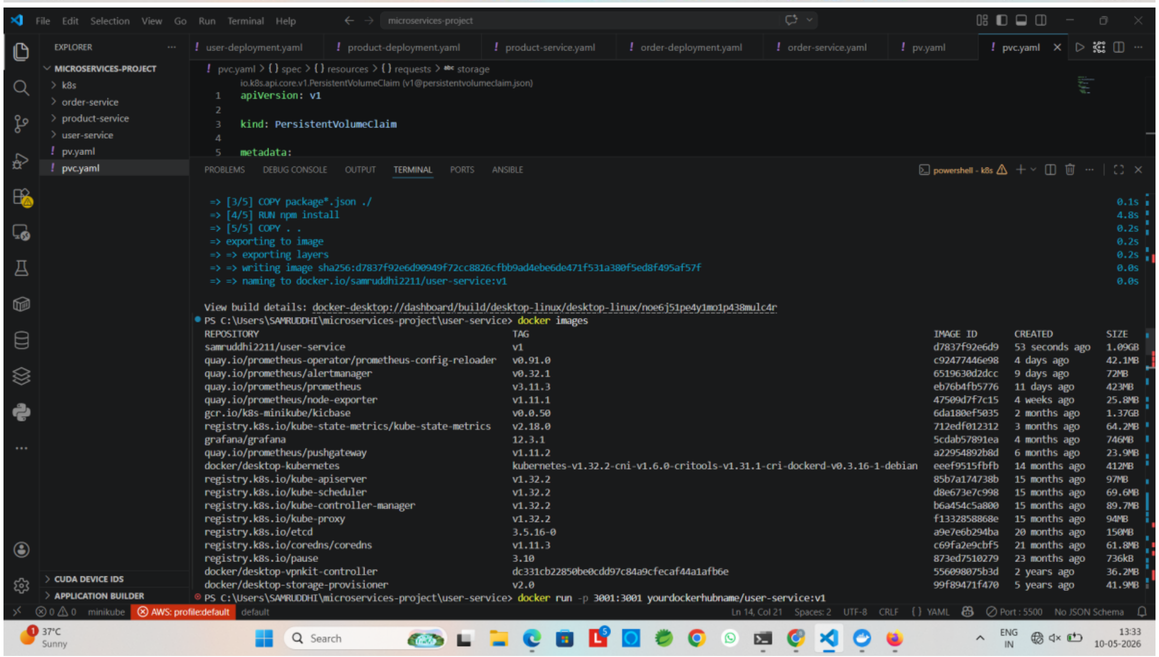
PS C:\Users\SAMRUDDHI\microservices-project\product-service> docker build -t samruddhi2211/product-service:v1 .
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/8ywi7d67jalfxprdfz9t9iybu
PS C:\Users\SAMRUDDHI\microservices-project\product-service> docker push samruddhi2211/product-service:v1
The push refers to repository [docker.io/samruddhi2211/product-service]
4369c84435f1: Pushed
4eb25f6511ba: Pushed
f7e7a42e06d6: Pushed
5e9016c8306b: Mounted from samruddhi2211/user-service
d2a991bcab4d: Mounted from samruddhi2211/user-service
b624aa2d5ea2: Mounted from samruddhi2211/user-service
d399c9dc306f: Mounted from samruddhi2211/user-service
84f9fa179c1b: Mounted from samruddhi2211/user-service
ce84ba212e49: Mounted from samruddhi2211/user-service
e4dc8cd9ecc8: Mounted from samruddhi2211/user-service
6428cc293366: Mounted from samruddhi2211/user-service
2f7436e79a0b: Mounted from samruddhi2211/user-service
v1: digest: sha256:d55a33812b07625611206102d3bbc86437cf8076872048f88055b7f884b25200 size: 2839
PS C:\Users\SAMRUDDHI\microservices-project\product-service> cd ..
PS C:\Users\SAMRUDDHI\microservices-project> cd order-service
PS C:\Users\SAMRUDDHI\microservices-project\order-service> npm init -y
Wrote to C:\Users\SAMRUDDHI\microservices-project\order-service\package.json:
{
"name": "order-service",
"version": "1.0.0",
"description": "",
"main": "index.js",
"scripts": {
"test": "echo \"Error: no test specified\" && exit 1"
},
"keywords": [],
"author": "",
"license": "ISC",
"type": "commonjs"
}
PS C:\Users\SAMRUDDHI\microservices-project\order-service> npm install express
added 65 packages, and audited 66 packages in 5s



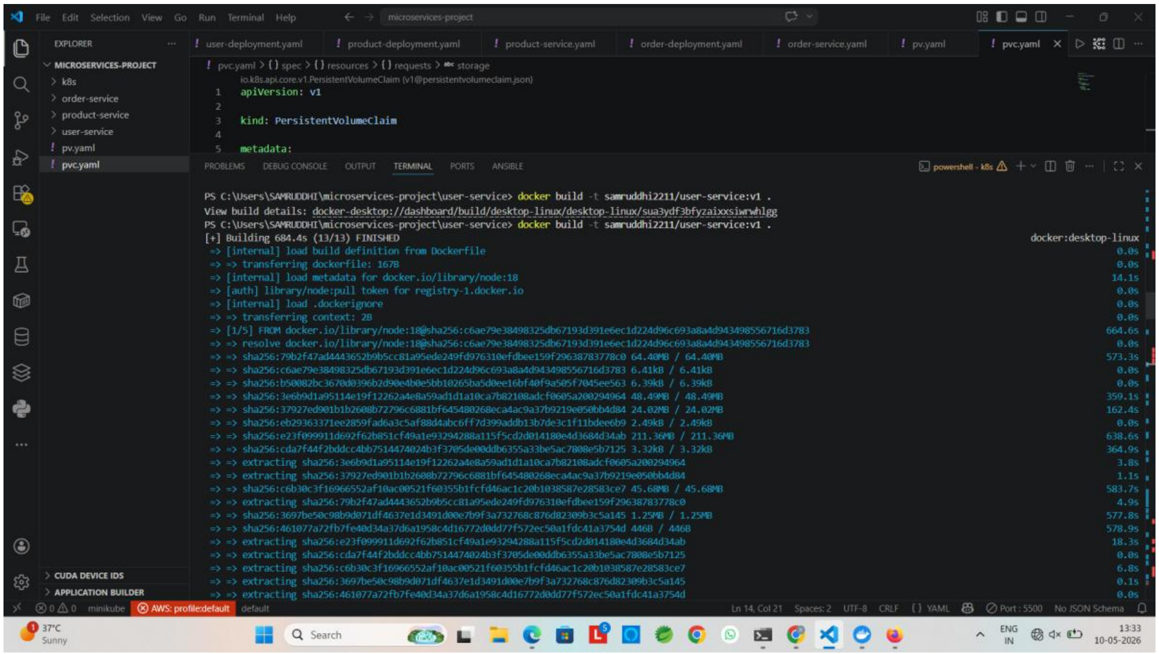
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
PS C:\Users\SAMRUDDHI\microservices-project\product-service> npm install express
added 65 packages, and audited 66 packages in 10s
22 packages are looking for funding
run `npm fund` for details
found 0 vulnerabilities
PS C:\Users\SAMRUDDHI\microservices-project\product-service> docker build -t samruddhi2211/product-service:v1 .
[+] Building 7.0s (11/11) FINISHED docker:desktop-linux
=> [internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 167B 0.0s
=> [internal] load metadata for docker.io/library/node:18 2.4s
=> [auth] library/node:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.1s
=> => transferring context: 2B 0.0s
=> [1/5] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4d943498556716d3783 0.0s
=> [internal] load build context 1.7s
=> => transferring context: 2.18MB 1.7s
=> CACHED [2/5] WORKDIR /app 0.0s
=> [3/5] COPY package*.json ./ 0.1s
=> [4/5] RUN npm install 2.3s
=> [5/5] COPY . . 0.2s
=> exporting to image 0.2s
=> => exporting layers 0.1s
=> => writing image sha256:834ca7094d558070ccb8c68751c175ce177d20068cc3bf412ccffc8837e6fda9 0.0s
=> => naming to docker.io/samruddhi2211/product-service:v1 0.0s
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/8ywi7d67jalfxprdfz9t9iybu
PS C:\Users\SAMRUDDHI\microservices-project\product-service> docker push samruddhi2211/product-service:v1
The push refers to repository [docker.io/samruddhi2211/product-service]
4369c84435f1: Pushed

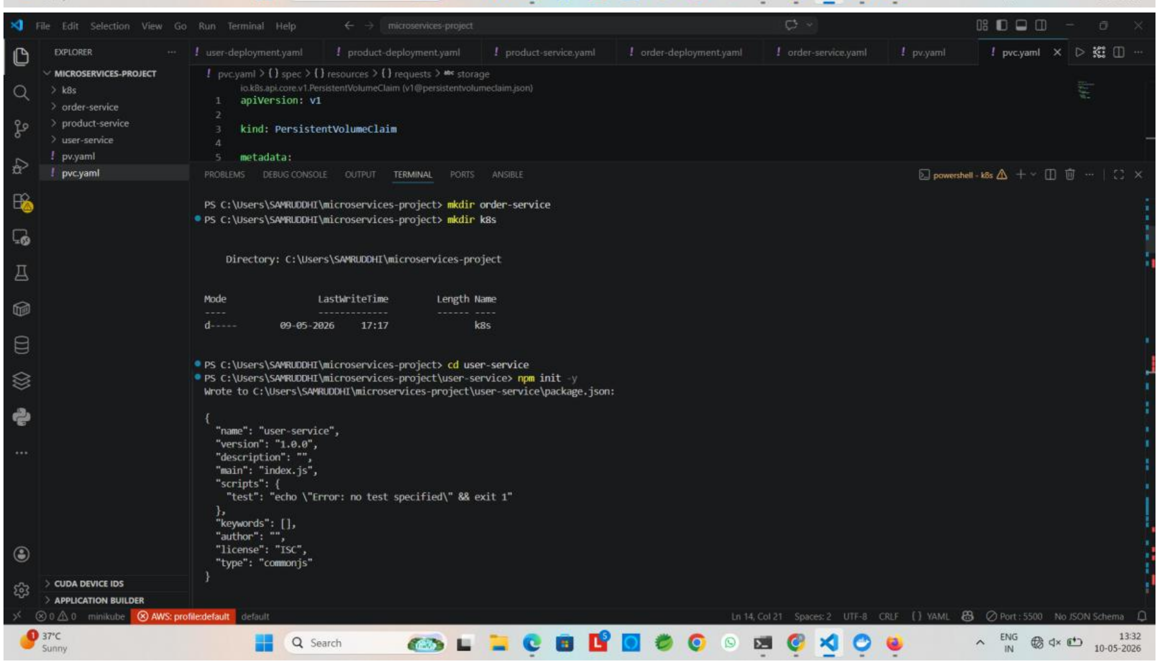


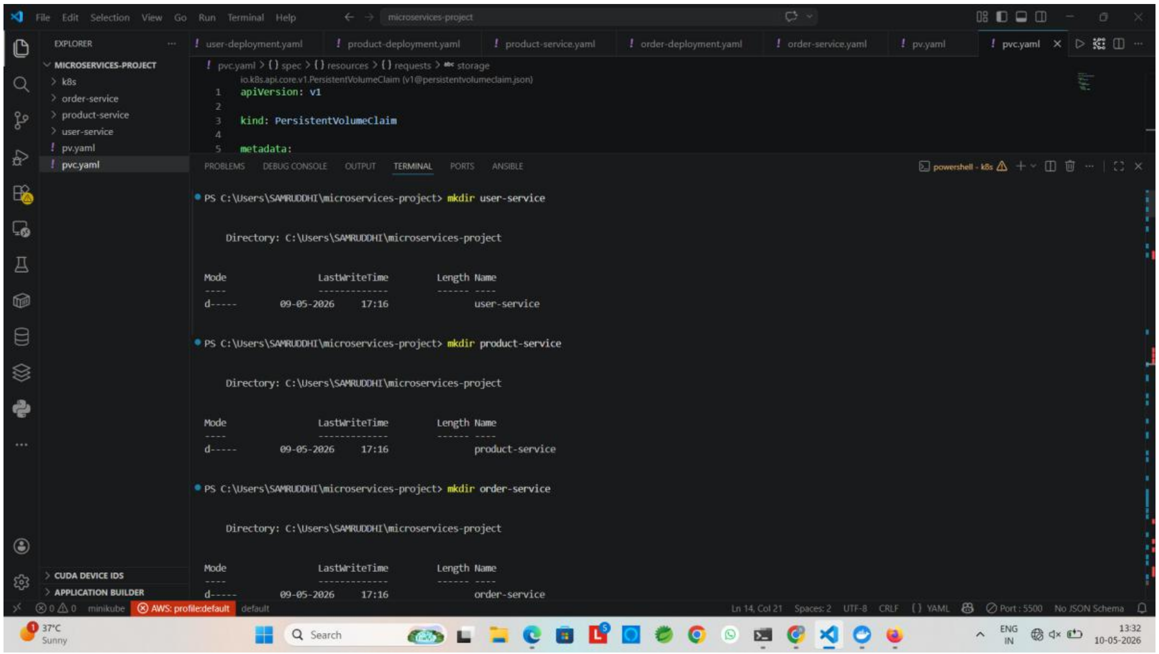
microservices-project
user-deployment.yaml
product-deployment.yaml
product-service.yaml
order-deployment.yaml
order-service.yaml
pv.yaml
pvc.yaml
EXPLORER
MICROSERVICES-PROJECT
k8s
order-service
product-service
user-service
pv.yaml
pvc.yaml
pvc.yaml > {} spec > {} resources > {} requests > storage
io.k8s.api.core.v1.PersistentVolumeClaim (v1@persistentvolumeclaim.json)
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
PROBLEMS
DEBUG CONSOLE
OUTPUT
TERMINAL
PORTS
ANSIBLE
powershell - k8s
PS C:\Users\SAMRUDDHI\microservices-project\user-service> docker run -p 3001:3001 yourdockerhubname/user-service:v1
PS C:\Users\SAMRUDDHI\microservices-project\user-service> docker run -p 3001:3001 samruddhi2211/user-service:v1
User Service Started
got 3 SIGTERM/SIGINTs, forcefully exiting
PS C:\Users\SAMRUDDHI\microservices-project\user-service> docker push samruddhi2211/user-service:v1
The push refers to repository [docker.io/samruddhi2211/user-service]
008638f861c7: Pushed
c9017ddb891b: Pushed
934acece12eb: Pushed
5e9016c8306b: Pushed
d2a991bcab4d: Mounted from library/node
b624aa2d5ea2: Mounted from library/node
d399c9dc306f: Mounted from library/node
84f9fa179c1b: Mounted from library/node
ce84ba212e49: Mounted from library/node
e4dc8cd9ecc8: Mounted from library/node
6428cc293366: Mounted from library/node
2f7436e79a0b: Mounted from library/node
v1: digest: sha256:65744d1bf06b59c0caa501ecabd12c6fec66b93bdb36bfb05978f43b8a41eb5a size: 2839
PS C:\Users\SAMRUDDHI\microservices-project\user-service> cd ..
PS C:\Users\SAMRUDDHI\microservices-project> cd product-service
PS C:\Users\SAMRUDDHI\microservices-project\product-service> npm init -y
Wrote to C:\Users\SAMRUDDHI\microservices-project\product-service\package.json:
{
"name": "product-service",
"version": "1.0.0",
"description": "",
"main": "index.js",
"scripts": {
CUDA DEVICE IDS
APPLICATION BUILDER
minikube
AWS: profile:default
default
Ln 14, Col 21
Spaces: 2
UTF-8
CRLF
{} YAML
Port : 5500
No JSON Schema
37°C
Sunny
Search
ENG
IN
13:34
10-05-2026

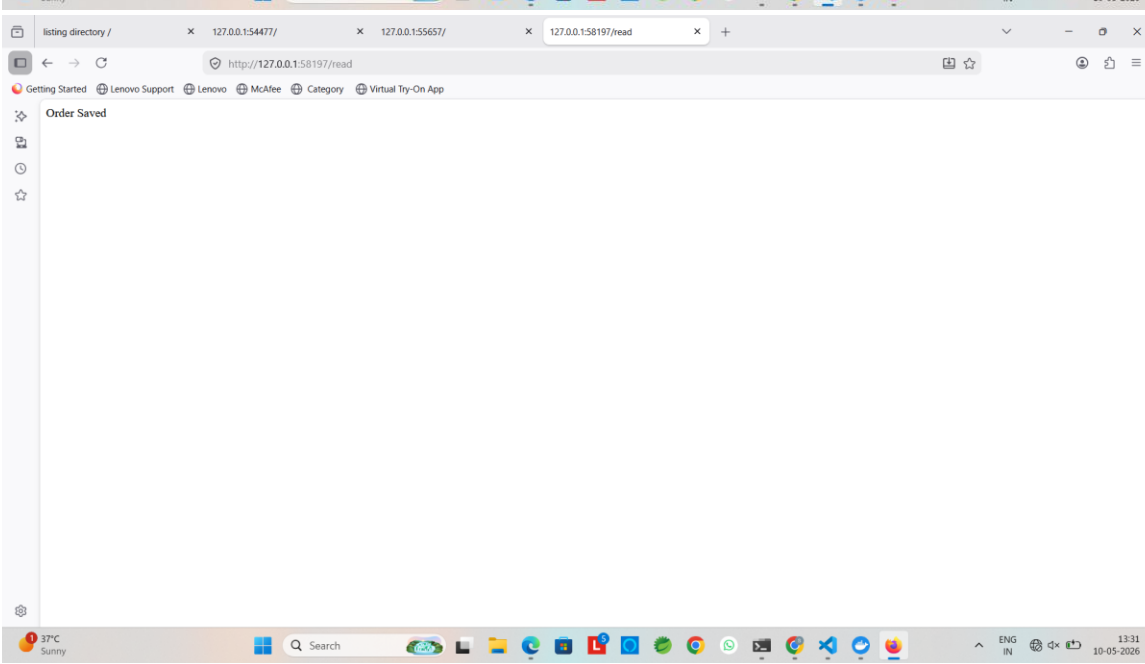


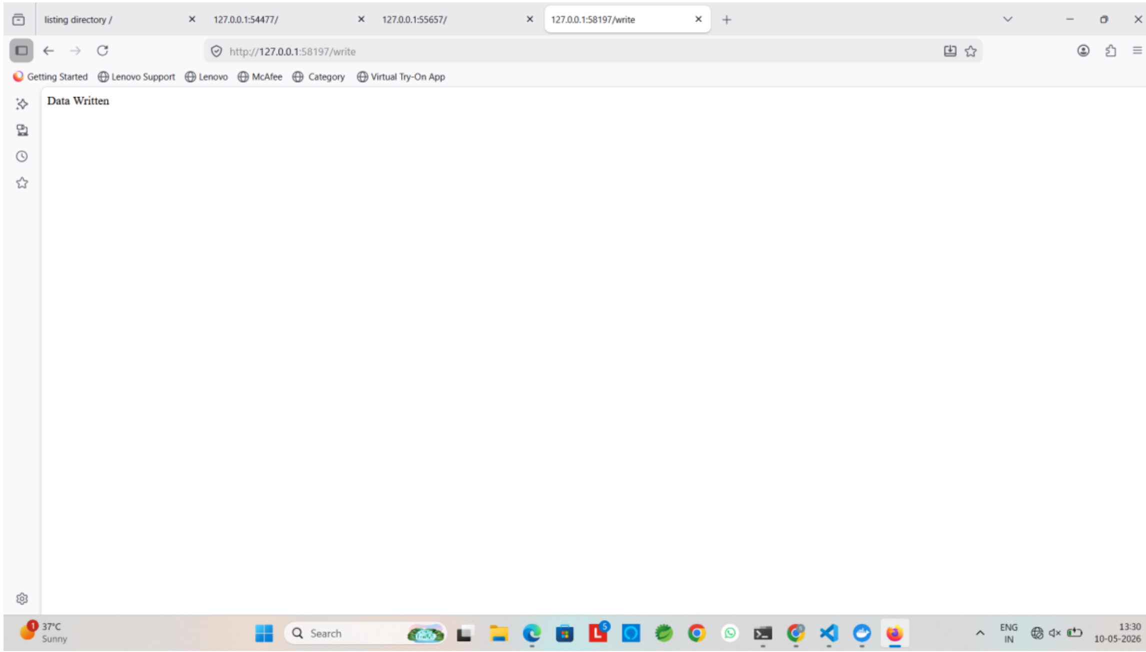
microservices-project
user-deployment.yaml
product-deployment.yaml
product-service.yaml
order-deployment.yaml
order-service.yaml
pv.yaml
pvc.yaml
EXPLORER
MICROSERVICES-PROJECT
k8s
order-service
product-service
user-service
pv.yaml
pvc.yaml
pvc.yaml > {} spec > {} resources > {} requests > storage
io.k8s.api.core.v1.PersistentVolumeClaim (v1@persistentvolumeclaim.json)
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
PROBLEMS
DEBUG CONSOLE
OUTPUT
TERMINAL
PORTS
ANSIBLE
powershell - k8s
=> [3/5] COPY package*.json ./ 0.1s
=> [4/5] RUN npm install 4.8s
=> [5/5] COPY . . 0.2s
=> exporting to image 0.2s
=> => exporting layers 0.2s
=> => writing image sha256:d7837f92e6d90949f72cc8826cfbb9ad4ebe6de471f531a380f5ed8f495af57f 0.0s
=> => naming to docker.io/samruddhi2211/user-service:v1 0.0s
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/noe6j51pe4y1mo1p438mulc4r
PS C:\Users\SAMRUDDHI\microservices-project\user-service> docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
samruddhi2211/user-service v1 d7837f92e6d9 53 seconds ago 1.09GB
quay.io/prometheus-operator/prometheus-config-reloader v0.91.0 c92477446e98 4 days ago 42.1MB
quay.io/prometheus/alertmanager v0.32.1 6519630d2dcc 9 days ago 72MB
quay.io/prometheus/prometheus v3.11.3 eb76b4fb5776 11 days ago 423MB
quay.io/prometheus/node-exporter v1.11.1 47509d7f7c15 4 weeks ago 25.8MB
gcr.io/k8s-minikube/kicbase v0.0.50 6da180ef5035 2 months ago 1.37GB
registry.k8s.io/kube-state-metrics/kube-state-metrics v2.18.0 712edf012312 3 months ago 64.2MB
grafana/grafana 12.3.1 5cdab57891ea 4 months ago 746MB
quay.io/prometheus/pushgateway v1.11.2 a22954892b8d 6 months ago 23.9MB
docker/desktop-kubernetes kubernetes-v1.32.2-cni-v1.6.0-critools-v1.31.1-cri-dockerd-v0.3.16-1-debian eeef9515fbfb 14 months ago 412MB
registry.k8s.io/kube-apiserver v1.32.2 85b7a174738b 15 months ago 97MB
registry.k8s.io/kube-scheduler v1.32.2 d8e673e7c998 15 months ago 69.6MB
registry.k8s.io/kube-controller-manager v1.32.2 b6a454c5a800 15 months ago 89.7MB
registry.k8s.io/kube-proxy v1.32.2 f1332858868e 15 months ago 94MB
registry.k8s.io/etcd 3.5.16-0 a9e7e6b294ba 20 months ago 150MB
registry.k8s.io/coredns/coredns v1.11.3 c69fa2e9cbf5 21 months ago 61.8MB
registry.k8s.io/pause 3.10 873ed7510279 23 months ago 736kB
docker/desktop-vpnkit-controller dc331cb22850be0cdd97c84a9cfecaf44a1afb6e 556098075b3d 2 years ago 36.2MB
docker/desktop-storage-provisioner v2.0 99f89471f470 5 years ago 41.9MB
PS C:\Users\SAMRUDDHI\microservices-project\user-service> docker run -p 3001:3001 yourdockerhubname/user-service:v1
CUDA DEVICE IDS
APPLICATION BUILDER
minikube
AWS: profile:default
default
Ln 14, Col 21
Spaces: 2
UTF-8
CRLF
{} YAML
Port : 5500
No JSON Schema
37°C
Sunny
Search
ENG
IN
13:33
10-05-2026

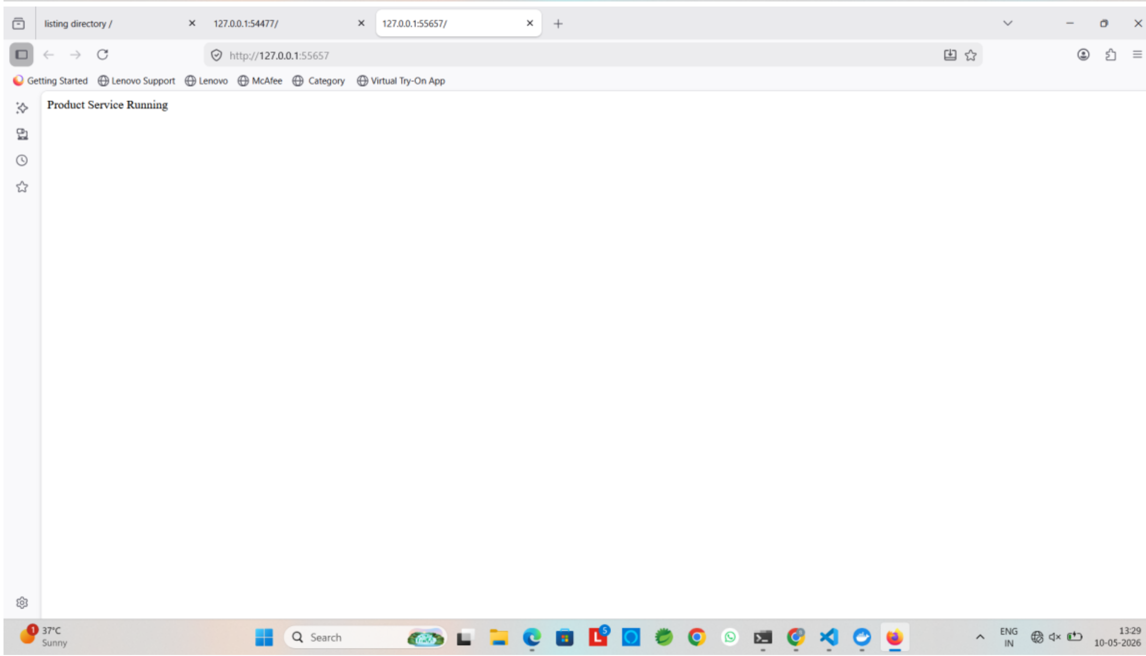

PS C:\Users\SAMRUDDHI\microservices-project\user-service> docker build -t samruddhi2211/user-service:v1 .
PS C:\Users\SAMRUDDHI\microservices-project\user-service> docker build -t samruddhi2211/user-service:v1 .
[+] Building 684.4s (13/13) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 167B
=> [internal] load metadata for docker.io/library/node:18
=> [auth] library/node:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B

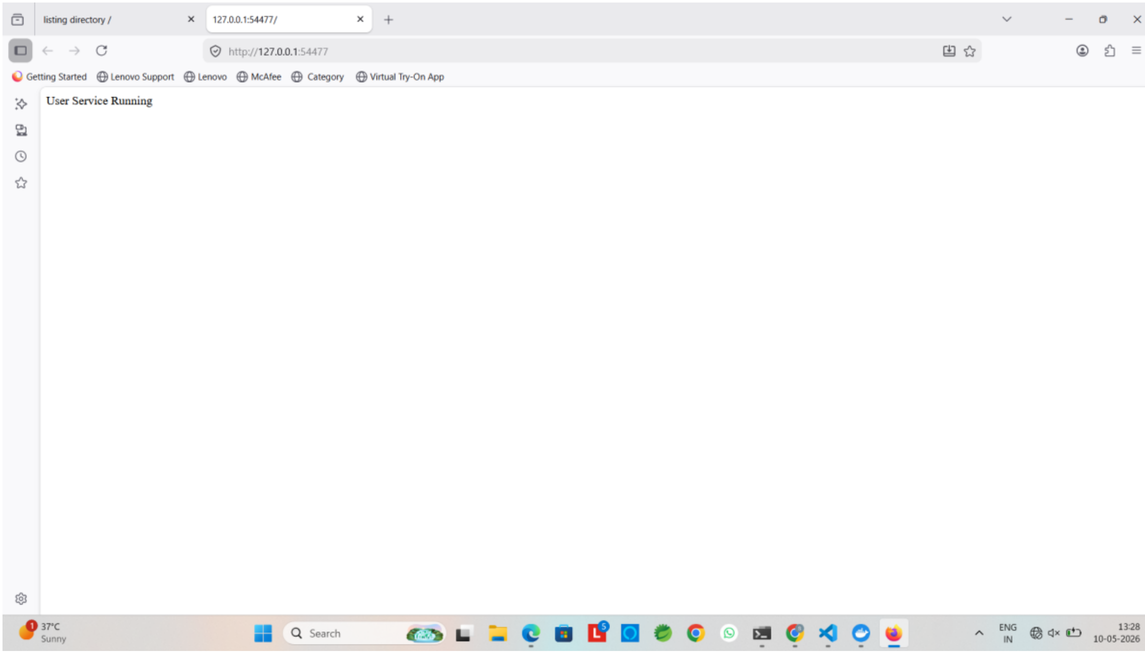

PS C:\Users\SAMRUDDHI\microservices-project> mkdir order-service
PS C:\Users\SAMRUDDHI\microservices-project> mkdir k8s
Directory: C:\Users\SAMRUDDHI\microservices-project
Mode LastWriteTime Length Name
d----- 09-05-2026 17:17 k8s
PS C:\Users\SAMRUDDHI\microservices-project> cd user-service
PS C:\Users\SAMRUDDHI\microservices-project\user-service> npm init -y
Wrote to C:\Users\SAMRUDDHI\microservices-project\user-service\package.json:
{
"name": "user-service",
"version": "1.0.0",
"description": "",
"main": "index.js",
"scripts": {
"test": "echo \"Error: no test specified\" && exit 1"
},
"keywords": [],
"author": "",
"license": "ISC",
"type": "commonjs"
}


File Edit Selection View Go Run Terminal Help
microservices-project
EXPLORER
MICROSERVICES-PROJECT
k8s
order-service
product-service
user-service
pv.yaml
pvc.yaml
user-deployment.yaml
product-deployment.yaml
product-service.yaml
order-deployment.yaml
order-service.yaml
pv.yaml
pvc.yaml
pvc.yaml > {} spec > {} resources > {} requests > storage
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
PROBLEMS DEBUG CONSOLE OUTPUT TERMINAL PORTS ANSIBLE
PS C:\Users\SAMRUDDHI\microservices-project> mkdir user-service
Directory: C:\Users\SAMRUDDHI\microservices-project
Mode LastWriteTime Length Name
d----- 09-05-2026 17:16 user-service
PS C:\Users\SAMRUDDHI\microservices-project> mkdir product-service
Directory: C:\Users\SAMRUDDHI\microservices-project
Mode LastWriteTime Length Name
d----- 09-05-2026 17:16 product-service
PS C:\Users\SAMRUDDHI\microservices-project> mkdir order-service
Directory: C:\Users\SAMRUDDHI\microservices-project
Mode LastWriteTime Length Name
d----- 09-05-2026 17:16 order-service
CUDA DEVICE IDS
APPLICATION BUILDER
minikube
AWS: profile:default
default
Ln 14, Col 21 Spaces: 2 UTF-8 CRLF {} YAML
Port : 5500 No JSON Schema
37°C
Sunny
Search
ENG
IN
13:32
10-05-2026


listing directory /
127.0.0.1:54477/
127.0.0.1:55657/
127.0.0.1:58197/read
http://127.0.0.1:58197/read
Getting Started
Lenovo Support
Lenovo
McAfee
Category
Virtual Try-On App
Order Saved
37°C
Sunny
Search
ENG
IN
13:31
10-05-2026


listing directory /
127.0.0.1:54477/
127.0.0.1:55657/
127.0.0.1:58197/write
http://127.0.0.1:58197/write
Getting Started
Lenovo Support
Lenovo
McAfee
Category
Virtual Try-On App
Data Written
37°C
Sunny
Search
ENG
IN
13:30
10-05-2026


listing directory /
127.0.0.1:54477/
127.0.0.1:55657/
http://127.0.0.1:55657
Getting Started
Lenovo Support
Lenovo
McAfee
Category
Virtual Try-On App
Product Service Running
37°C
Sunny
Search
ENG
IN
13:29
10-05-2026



14. Commands Summary

- minikube start
- kubectl get nodes
- docker build -t username/service:v1 .
- docker push username/service:v1
- kubectl apply -f deployment.yaml
- kubectl get pods
- kubectl get svc
- kubectl get hpa
- kubectl describe pod <pod-name>